DESIGN SYSTEMS

TECHNICAL DATA SHEET EPOX 645
TWO-COMPONENT EPOXY FINISH

DESCRIPTION

Two-component, water-dilutable, transparent, scratch-resistant epoxy finish, resistant to UV rays.

It makes the substrate washable and easy to sanitise.

GENERAL CHARACTERISTICS

Epox 645 is a latest-generation two-component product based on modified, nanostructured, water-soluble, low-viscosity epoxy resins.

It has good adhesion to a wide range of substrates such as cement, micro-cement, stone, iron, steel, sheet metal and wood; it has good resistance to acids and organic solvents, excellent resistance to alkalis, good resistance to abrasion and traffic, good breathability and excellent light resistance; ideal for a hygienic environment, VOC release = 0.

FIELDS OF APPLICATION

Epox 645 is a specific enamel for the protection and transparent finishing of cement floors, resin floors, wood laminates and wooden items such as doors, windows, furniture, etc.

The product is suitable as a protective finish for swimming pools and tanks.

PHYSICAL PROPERTIES (Typical Values)

Colour	Milky white
Appearance	Liquid
Version	Glossy and matt
Brookfield viscosity at 25°C	500-900 cps
Density at 20°C, glossy version	1.05 kg/l ± 0.05
Mixing ratio A : B (A = base, B = hardener)	100 : 33
Mixing ratio A : B (A = base, B = hardener), matt version	100 : 37
Pot life (product diluted 10%) at 20°C	40 minutes
Dust-free time at 20°C	3 hours
Overcoating time at 20°C	8 hours
Curing time at 20°C	12 hours
Foot traffic	24 hours
Time to full service	4 days
Application temperature	from +7°C to +35°C

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Flammability	non-flammable
Recommended dry film thickness	40-50 µm

PREPARATION

Before applying the product, the surface must be clean, dry, free from oil, grease, paint residues and other removable parts. Remove every trace of any contaminant. Substrates must be lightly abraded mechanically and must be free from insoluble salts on the surface. Once applied, the product is milky white in colour and becomes increasingly transparent as it dries. Avoid application in unsuitable ambient conditions and in any case at temperatures below +7 °C.

APPLICATION

Application can be carried out by brush, roller or spray.

Important Note

The quality of the application will be influenced by the equipment and tools used, as well as by the ambient conditions during application, such as temperature, relative humidity and the presence of dust in the application area.

When applying the product for the first time, it is recommended to carry out some preliminary tests in order to check and validate the process and ensure the best technical and aesthetic result.

Apply the product in ambient conditions with a temperature between +7 °C and +35 °C and a maximum relative humidity of 70%.

When opening each pack, stir the product thoroughly to eliminate any sediment.

Mix the components well in the indicated ratio.

Dilution:

For the first coat, dilute with drinking water by 40 to 50% max; for the second coat, dilute by up to 35%.

The life of the catalysed product, i.e. the time available for application, is approximately 40 minutes (at ambient conditions); depending on the application, it is recommended to prepare a suitable quantity of product so as to ensure optimal workability.

Drying reaches the “dust-free” stage in 3 hours at 20 °C, overcoating in 8 hours at 20 °C, full curing in 12 hours at 20 °C.

The theoretical coverage depends on the type of substrate to be treated. On average, 1 kg of product (A+B) treats 8-10 m^2 of surface.

PACKAGING

Plastic drums of:

GLOSSY VERSION	MATT VERSION
5.400 kg (part A) + 2.000 kg (part B)	6.000 kg (part A) + 2.000 kg (part B)

1.350 kg (part A) + 0.500 kg (part B)	3.000 kg (part A) + 1.000 kg (part B)
	0.900 kg (part A) + 0.300 kg (part B)

STORAGE

Protect from frost, heat and direct sunlight. In original, intact containers, protected from frost and excessive heat, the product keeps for approximately 12 months.

STORAGE

Do not eat, drink or smoke during application. Use personal protective equipment: gloves, goggles, mask and protective clothing of the type indicated in the safety data sheet. The information contained in this data sheet is, to the best of our knowledge, correct and accurate, but every recommendation and suggestion given is without any guarantee, since the conditions of use are not under our direct control. In case of doubt, it is always advisable to carry out preliminary tests.

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Always comply with the regulations and provisions in force regarding safety and transport. Consult the technical data sheet, the application guidelines and the safety data sheet. For any clarification or information, please contact our Technical Customer Service.

The technical data given in this bulletin were obtained using standardised tests and standards, or modifications thereof, according to Design Systems systems. The methods applied may be requested from our technical service. The contents of this technical bulletin represent the best of our knowledge and experience. However, for uses other than those indicated, preliminary tests are recommended. In any case, Design Systems Srl accepts no responsibility for damage or defects caused by the use of its products, since the conditions of use are not under its control. It also advises that its technical service is available to customers for any information concerning the correct use of its products.